

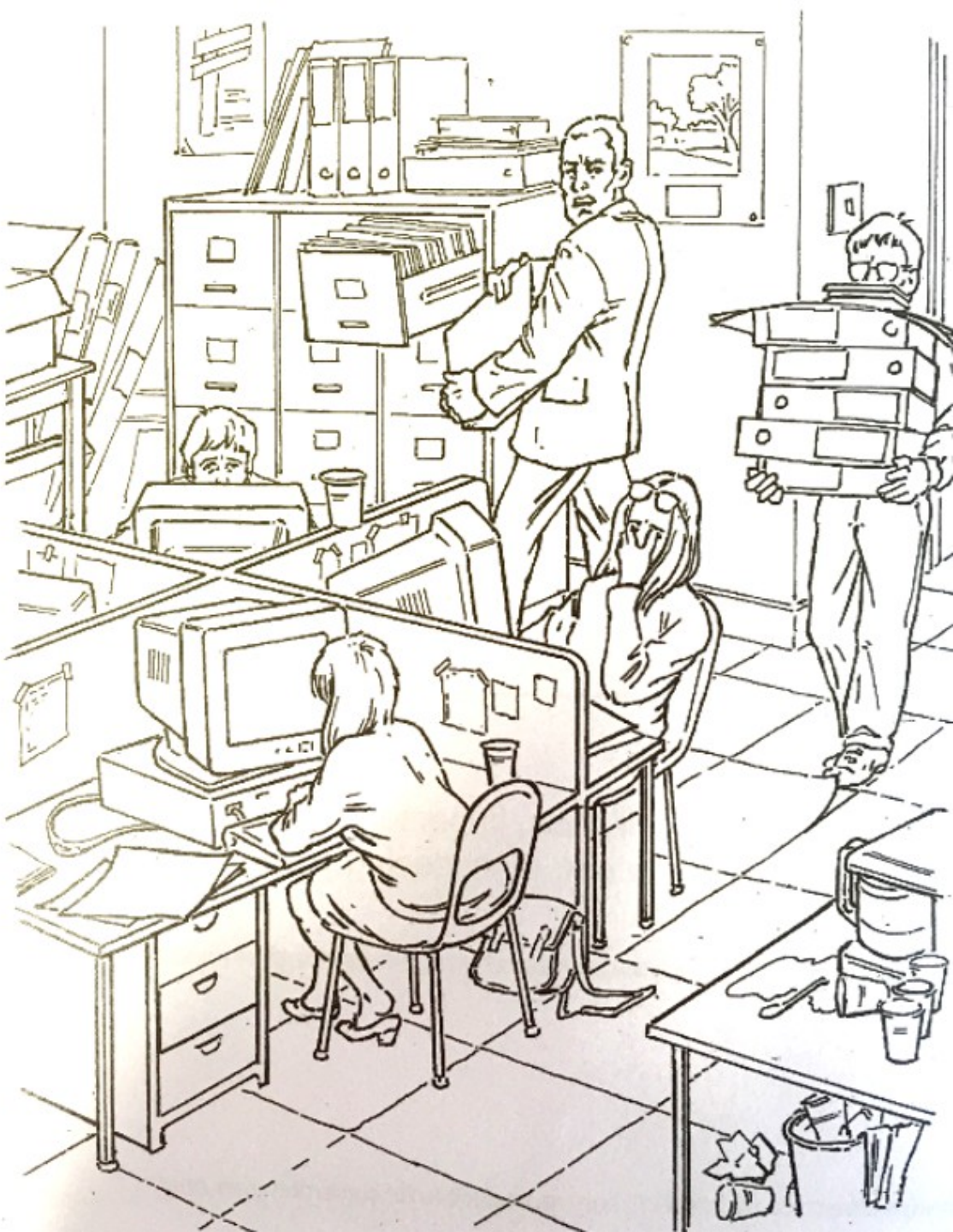
Safety Using ICT

Mr Butler | Unit 2

There are 10 health hazards in this office.

*Can you identify them?

CHALLENGE: Explain why they are hazards?



Learning Objectives

- Know the importance of health and safety when using technology
- Appreciate how **common sense** can help protect our workplace
- Learn some rules about how to keep **data safe**
- Appreciate the importance of protecting our data, and ourselves when using ICT



Working Safely

- When using computers, we need to keep certain things safe:
  - The equipment
  - Our data
  - Ourselves



- We need to follow **Health and Safety rules** to keep ourselves safe from any harm.
- Most of the rules are common sense – we just need to behave sensibly.
- If something looks dangerous, report it to the person in charge.

Back Against the Wall

Failing to stick to health and safety rules can result in injury



Back Against the Wall

- **Backache** can be caused by sitting for long periods of time in the same position and having your chair and the wrong height. There is also the possibility of damage to the spine being caused if this is not addressed.



Back Against the Wall

- To Combat this potential threat:
  - Use an adjustable chair
  - Take regular breaks
  - Use a footrest
  - Change your sitting position (don’t slouch)



A Sight for Sore Eyes

- **Eye strain** is caused by continuously looking at a screen for an extended period of time. Reflection and glare from bright lights can make it worse.
- Constantly straining your eyes can result in headaches, sore eyes and a worsening of your eyesight.



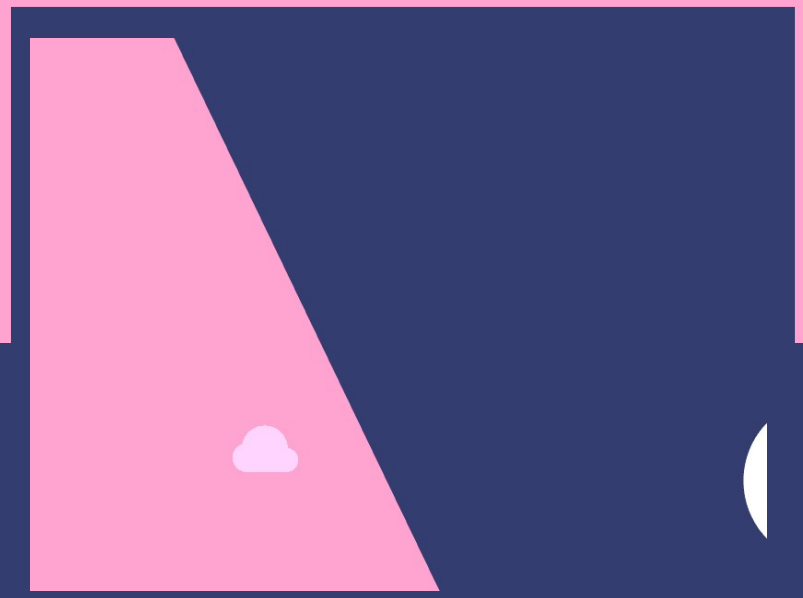
A Sight for Sore Eyes

- To combat this potential threat:
  - Use suitable screen size
  - Take regular breaks
  - Fit blinds to windows
  - Position your monitor away from direct sunlight
  - Use an anti-glare screen



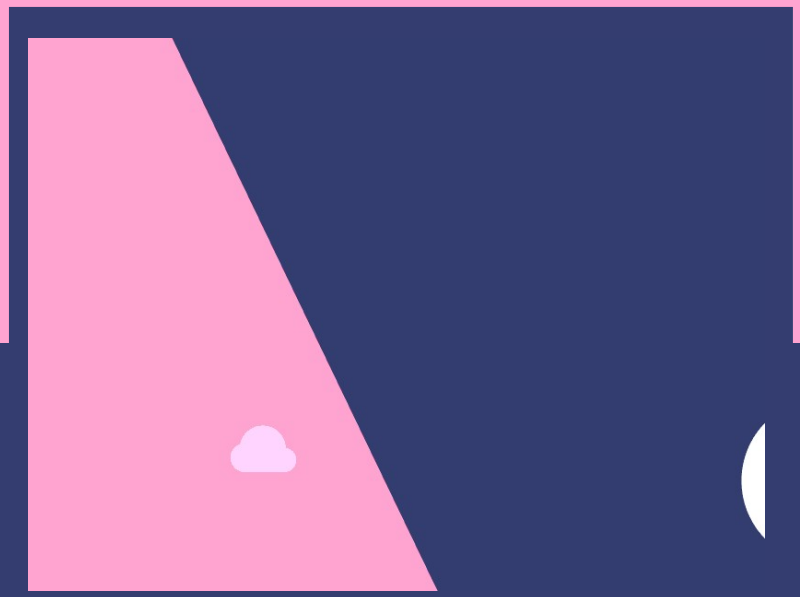
Day After Day After Day

- **Repetitive Strain Injury (RSI)** happens when you constantly make the same movement again and again. It usually affects joints in wrists and fingers.
  - Ever played a game for a long time and got sore/numb thumbs? This is has been termed **Nintendo thumb.**
- It can cause extreme pain and damage to the joints – in extreme cases resulting in arthritis.



Day After Day After Day

- Combating this potential threat to your health, we can:
  - Adjust your keyboard to a comfortable angle.
  - Use a wrist rest
  - Keep wrists straight when typing
  - Take regular breaks





Hazard

What is a Hazard?



Discuss this with the person next to you to come up with a definition.



Definition: Hazard

A danger or risk which is any source of **potential** damage, harm or adverse health effects on something or someone.

What is Ergonomics?

- The science concerned with designing safe and comfortable hardware and office furniture for humans
- Ergonomics is designing equipment suitable for humans – for example:
  - PlayStation has officially altered its PS4 controller 3 times, although there are hundreds more on the market!
  - Microsoft spent “hundreds of millions” designing its xBox One controller

Homework: Due Next Lesson

- Ensure you write your homework into your planner.
- List 5 hazards and possible ways to combat these hazards in the ICT classroom.
- Place this into a Microsoft Word document and upload this **before** next lesson.

Worksheet: Ergonomics



- Using what you have learnt today, complete the ‘Ergonomics’ worksheet.

Cyber Security



Call Security!



- As well as keeping ourselves protected when using ICT, we need to protect our data from being damaged.
- There are many different data threats:
  - Unauthorised users, or **hackers**, can gain access to data by illegal means.
  - **Viruses** can attack and seriously damage files.
  - **System crashes** which affect the hard drive may damage key sections of data.
  - Data can be lost **accidentally** or deleted by mistake.
  - Data can also be **sabotaged** by former employees who are guilty of committing **malicious acts**.

Good use of equipment to avoid ICT related injury



Worksheet: Every Problem has a Solution

Using your notes, complete the

‘Every Problem has a Solution’

Worksheet.



Keeping Data Safe

- Answer these questions on a Microsoft Word document called '**Keeping Data Safe**' by offering a solution to these problems:

1. Saved files cannot be found
2. Computer is damaged
3. File has a virus
4. Forgotten the file name
5. Password goes missing/forgotten



Research Task



- In a **creative** way, state 8 employer regulations that they must follow to keep their employees safe from ICT related injury.

  - Use Colour
  - Use appropriate fonts
  - Images/diagrams
  - Styling (including borders/page colour/etc)

https://www.bbc.co.uk/bitesize/guides/z7t6jhv/revision/1

Worksheet: Word Search

Complete the word search worksheet.

Use a different colour for each word!



Homework - Situation



Complete the worksheet ‘Situations’ and decide what is the best course of action for each set of circumstances.

Then complete the research task at the bottom of the sheet.